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MILITARY SPECIFICATION

SHEARING MACHINE, METAL SQUARING, MECHANICAL DRIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanically driven, metal squaring shearing machines with equipment and accessories specified herein.

1.2 Classification. The shearing machines shall be of the sizes listed in table I. The size to be furnished shall be as specified (see 6.2.1 and table I).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standard.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.4 - Safety Requirements for the Construction, Care, and Use of Shears.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the Publication Sales Division, National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The shearing machine shall be new and one of the manufacturer's current models capable of shearing sheet metal through the downward movement of a ram mounted upper shear knife past a table mounted stationary shear knife. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the shearing machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, front or back gauge indicators, dials, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1).

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Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The shearing machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The shearing machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, and other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the applicable requirements contained in ANSI B.11.4 and CFR 29, Chapter XVII, Part 1910. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The shearing machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the shearing machine.

3.2.8 Environmental protection. The shearing machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation (see 6.4).

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3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Each lubricant reservoir shall have a means of determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible. Unless otherwise specified (see 6.2.1), machines of size 3/16 inch capacity and smaller shall be provided with either a manual one-shot centralized pressure lubrication system or an automatic centralized lubrication system. Machines of sizes 1/4 inch capacity and larger shall be provided with an automatic centralized lubrication system.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The shearing machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

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3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a frame, a drive system, a holddown system, gauges, and shear knives.

3.4.1 Frame. The frame assembly shall support all other components of the machine at a level of rigidity required to successfully perform the shearing operations specified herein. The frame shall consist of two end housings, a table, and a ram joined together with a bolted, interlocking construction.

3.4.1.1 End housings. The end housings shall be either the closed end or gap type design. When only one type end housing design is acceptable the particular type shall be as specified (see 6.2.1). When a gap type end housing design is furnished, the gap shall be recessed not less than 6 inches from the lower shear knife on size 3/16 inch and smaller machines and not less than 18 inches on size 1/4 inch and larger machines. The base of the end housings shall have provisions for securely anchoring the machine to a flat mounting foundation.

3.4.1.2 Table. The table shall be constructed of either cast iron or steel plate with a surface finish of 125 microinches or less, and flat to within a tolerance of +0.005 inch per linear foot. The table shall securely support the lower shear knife and shall be provided with hand slots to facilitate workpiece handling. The surface of the table shall also be provided with slots, as required, for attaching any workpiece positioning gauges specified herein. The table shall be provided with a means of adjustment for obtaining and maintaining clearance alignment between the upper and lower shear knives.

3.4.1.3 Ram. The ram shall be constructed of either cast iron or steel plate and shall secure and support the upper shear knife. The stroke of the ram shall be accomplished on adjustable guide bearings capable of maintaining the shearing accuracy specified herein. A knife seat shall be machined across the full width of the ram for directly supporting the upper shear knife in an inclined position to comply with the rake angle requirements contained herein.

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3.4.2 Drive system. The machine shall have an electric motor driven mechanical drive system capable of initiating and controlling the reciprocating stroke of the ram. A dynamically balanced cast iron or steel flywheel shall supply supplemental power to the drive system. An eccentric shaft shall convert the rotary motion of the motor and flywheel into the reciprocating motion required by the ram to complete the shearing stroke. The eccentric shaft shall rotate within bushings or anti-friction bearings and shall be directly supported by the end housings of the frame. The energy developed by the drive system shall be controlled by a clutch capable of transmitting the maximum developed torque to the eccentric shaft and ram. The clutch shall be provided with a brake for stopping the motion of the ram at the top of the ram stroke when the control is set for single stroke operation.

3.4.2.1 Drive system operating controls. The machine shall be provided with controls for operating the ram in both single stroke and continuous (repeating) stroke operation. The drive system shall be operated by either a treadle or foot switch type control.

3.4.3 Holddown system. The machine shall be provided with a holddown system capable of securing any size workpiece up to and including the rated capacity of the machine. The holddown system shall consist of either spring loaded, hydraulic, or pneumatic type plunger units, with a centerline spacing of not more than 12 inches across the full workpiece opening of the machine. The plunger units shall apply uniform holddown pressure, preventing any workpiece movement that would impede the required shearing performance specified herein. The plunger units shall automatically secure the workpiece to the machine table during the ram downstroke and release the workpiece on the return stroke.

3.4.4 Gauges. The machine shall be provided with a back gauge, a front gauge, and two side squaring gauges. The gauges shall accommodate any workpiece width up to and including the rated capacity of the machine.

3.4.4.1 Back gauge. The back gauge shall provide an adjustable reference edge between the back edge of the workpiece and the lower shear knife. The back gauge shall be adjustable to any position within the back gauge capacity specified herein. Size 10 gauge and smaller capacity machines shall be furnished with either a power or manually operated back gauge. Size 3/16 inch and larger machines shall be provided with a power operated back gauge. The operating controls of both the power and manually operated back gauges shall be located on the front of the machine and shall be provided with a direct reading counter type position indicator that displays the distance between the back gauge and the lower shear knife. A means of adjustment shall be provided for correcting any back gauge misalignment. When specified (see 6.2.1), the back gauge shall be furnished with either a magnetic or pneumatic workpiece support mechanism for supporting the off-cut portion of capacity width workpiece sizes of 16 gauge material and less. A means of adjustment shall be provided for correcting any misalignment of the back gauge.

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3.4.4.2 Front gauge. The front gauge shall consist of not less than two detachable support arms with adjustable work stops, to provide an adjustable reference edge between the front edge of the workpiece and the lower shear knife. The work stops shall lock in position throughout the full length of the support arms.

3.4.4.3 Side squaring gauge. Both the right and left side of the machine shall be provided with a detachable side squaring gauge for positioning the edge of the workpiece at right angles to the lower shear knife. Graduated scales to facilitate workpiece positioning shall be recessed in either the side squaring gauges or the surface of the bed. When specified (see 6.2.1), an extension side squaring gauge with length as specified shall be provided with the machine. The extension side squaring gauge shall be provided with a graduated scale and an adjustable workstop.

3.4.4.4 Light beam shearing gauge. When specified (see 6.2.1), the machine shall be provided with a light beam shearing gauge that utilizes light beam shadows at the shear line to facilitate shearing to a scribed line on the workpiece.

3.4.5 Shear knives. Shearing machines of 1/4 inch capacity and less shall be provided with high carbon, high chrome tool steel shear knives. Shearing machines of 3/8 inch capacity and greater shall be provided with shock resisting tool steel shear knives. The shear knives shall be reversible, shall have four cutting edges, and shall be hardened and ground to comply with the manufacturer's standard practice. When specified (see 6.2.1), an extra set of shear knives shall be furnished with the machine.

3.4.6 Electrical system. The electrical system shall conform to NFPA 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.6.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure.

3.4.6.2 Solid state components. Solid state design shall be used throughout the machine for electronic components.

3.4.6.3 Control circuit voltage. Control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

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3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of shearing sheet steel with a tensile strength of not less than 60,000 psi in workpiece sizes ranging up to the rated capacity for the thickness and width as specified in table I. The sheared edge of the workpiece shall be within 3 degrees of perpendicular with the flat side of the workpiece. The workpiece shall be free from holddown damage.

3.6.1 Shearing accuracy. The machine shall be capable of shearing 16 gauge material ranging up to the full width capacity of the machine to the following tolerances.

a. Dimensions - The width of the off-cut portion of the workpiece shall be within ± 0.005 inch of the back gauge setting.

b. Parallelism - The sheared cut shall be parallel to the sheared (trimmed) edge positioned against the backgauge to within ± 0.002 inch per linear foot of workpiece width.

c. Squareness - The sheared cut shall be square with a sheared (trimmed) edge positioned against either side gauge to within ± 0.002 inch per linear foot of workpiece width.

3.6.2 Additional performance requirements. Performance features or shearing accuracies required in addition to or in lieu of the requirements contained herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. All equipment normally furnished with the manufacturer's standard commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets with suitable means of mounting. The chart or plate shall contain the following information:

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TABLE I. Sizes and capacities.

Machine size	Workpiece thickness	Workpiece width (feet)	Rake angle (inches per foot)	Shearing speed	Back gauge capacity (inches)	Front gauge capacity (inches)
16 ga. x 3	16 gauge	3	3/8	60	24	32
16 ga. x 4	16 gauge	4	3/8	60	24	32
16 ga. x 5	16 gauge	5	3/8	60	24	32
16 ga. x 6	16 gauge	6	5/16	60	24	32
16 ga. x 12	16 gauge	12	1/4	60	24	36
14 ga. x 3	14 gauge	3	13/32	60	24	32
14 ga. x 4	14 gauge	4	13/32	60	24	32
14 ga. x 5	14 gauge	5	13/32	60	24	32
14 ga. x 6	14 gauge	6	13/32	60	24	32
14 ga. x 8	14 gauge	8	3/8	60	24	36
14 ga. x 10	14 gauge	10	3/8	60	24	36
14 ga. x 12	14 gauge	12	3/8	60	24	36
12 ga. x 4	12 gauge	4	3/8	60	24	36
12 ga. x 6	12 gauge	6	3/8	60	24	36
12 ga. x 8	12 gauge	8	3/8	60	24	36
12 ga. x 10	12 gauge	10	3/8	60	24	36
12 ga. x 12	12 gauge	12	3/8	60	24	36
12 ga. x 14	12 gauge	14	3/16	60	36	50
10 ga. x 4	10 gauge	4	3/16	60	24	36
10 ga. x 6	10 gauge	6	3/8	60	24	36
10 ga. x 8	10 gauge	8	3/8	60	24	36
10 ga. x 10	10 gauge	10	3/8	60	24	41
10 ga. x 12	10 gauge	12	3/8	60	24	41
10 ga. x 14	10 gauge	14	1/4	60	24	41
10 ga. x 16	10 gauge	16	1/4	60	36	50

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TABLE I. Sizes and capacities - Continued.

Machine size	Workpiece thickness	Workpiece width (feet)	Rake angle (inches per foot)	Shearing speed	Back gauge capacity (inches)	Front gauge capacity (inches)
3/16 x 4	3/16 inch	4	7/16	60	24	36
3/16 x 6	3/16 inch	6	3/8	60	24	36
3/16 x 8	3/16 inch	8	3/8	60	24	36
3/16 x 10	3/16 inch	10	3/8	60	24	41
3/16 x 12	3/16 inch	12	3/8	55	24	41
3/16 x 14	3/16 inch	14	15/64	55	36	50
3/16 x 16	3/16 inch	16	7/32	50	36	54
3/16 x 18	3/16 inch	18	7/32	50	36	56
1/4 x 4	1/4 inch	4	3/8	60	24	41
1/4 x 6	1/4 inch	6	3/8	60	24	41
1/4 x 8	1/4 inch	8	3/8	60	36	41
1/4 x 10	1/4 inch	10	3/8	60	36	41
1/4 x 12	1/4 inch	12	3/8	60	36	41
1/4 x 14	1/4 inch	14	5/16	50	36	45
1/4 x 20	1/4 inch	20	1/4	45	36	53
3/8 x 4	3/8 inch	4	11/16	50	36	48
3/8 x 6	3/8 inch	6	1/2	50	36	48
3/8 x 8	3/8 inch	8	5/16	50	36	48
3/8 x 10	3/8 inch	10	5/16	50	36	48
3/8 x 12	3/8 inch	12	5/16	50	36	48
3/8 x 16	3/8 inch	16	5/16	45	36	48
3/8 x 20	3/8 inch	20	5/16	36	36	48
1/2 x 4	1/2 inch	4	7/16	50	36	48
1/2 x 6	1/2 inch	6	1/2	50	36	48
1/2 x 8	1/2 inch	8	13/32	40	36	48
1/2 x 10	1/2 inch	10	13/32	40	36	48
1/2 x 12	1/2 inch	12	13/32	40	36	48

Requirements shall be not less than the figures shown except for the rake angle requirements which shall be not more than the figure shown.

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Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. The machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes, during which time the drive, holddown system, and power operated gauges shall be actuated and observed to determine compliance with the requirements specified herein.

4.7.2 Performance test. The machine performance shall be tested by operating the machine to shear sheet steel with a tensile strength of not less than 60,000 psi. The instruments and methods used in performing the test

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shall be at the option of the manufacturer, provided they determine the capacity and gauge shearing capabilities of the machine.

4.7.2.1 Capacity shearing test. Capacity shearing shall be verified by performing shear cuts in the rated workpiece size and shearing speed capacities specified in table I as applicable for the size machine specified in the contract or order. The shear cuts shall comply with 3.6.

4.7.2.2 Shearing accuracy test. Shearing accuracy shall be verified by utilizing the back gauge and side squaring gauge to perform parallel and square shearing on 16 gauge workpieces of not less than the full width capacity, as applicable for the size machine specified. Shearing accuracy shall comply with 3.6.1.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The shearing machines covered by this specification are intended for use in metal shearing and squaring operations that are generally required in all phases of metal fabrication work.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific dimensional system (US or SI), state required system (3.2.1).

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e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Lubrication system, if different (see 3.2.9).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

k. If a particular type end housing design is required, state required design (see 3.4.1.1).

l. Workpiece support mechanism, if required (see 3.4.4.1).

m. Extension side squaring gauge, if required specify length (see 3.4.4.3).

n. Light beam shearing gauge, if required (see 3.4.4.4).

o. Extra set of shear knives, if required (see 3.4.5).

p. Motor enclosure, if different (see 3.4.6.1).

q. Size and capacity, if different (see 3.5).

r. Additional performance features or shearing accuracies, if required (see 3.6.2).

s. Optional equipment, if required (see 3.8).

t. Specify applicable repair parts, if required (see 3.9).

u. First article inspection, if different (see 4.4).

v. Quality conformance inspection, if different (see 4.5).

w. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt

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plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machine in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (g).

6.5 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification are as follows:

The types, classes, and styles previously covered by 00-S-305 have been deleted. MIL-S-80271 allows for the competitive procurement of both the closed end and open end housing type shearing machine on an equal basis. The following sizes have been deleted or added as indicated:

Deleted	Added
10 gauge x 24 inch	12 gauge x 14 feet
10 gauge x 36 inch	
10 gauge x 60 inch	
3/16 inch x 24 inch	
3/16 inch x 36 inch	
1/4 inch x 216 inch	
3/8 inch x 168 inch	
3/8 inch x 216 inch	

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Deleted

16 gauge x 42 inch
16 gauge x 96 inch
16 gauge x 120 inch
16 gauge x 216 inch

14 gauge x 42 inch

12 gauge x 24 inch
12 gauge x 36 inch
12 gauge x 60 inch

5/8 inch x 48 inch
5/8 inch x 72 inch
5/8 inch x 168 inch
5/8 inch x 142 inch

3/4 inch x 96 inch
3/4 inch x 120 inch
3/4 inch x 144 inch

1 inch x 48 inch
1 inch x 72 inch

6.6 Subject term (key word) listing.

Cutting
Metal cutting
Metal working
Shearing
Slicing
Slitting
Trimming

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - YD
Air Force - 84
DLA- GS

User activities:
Army - AV
Navy - MC, OS

Preparing activity:
DLA - IP

(Project 3445-0098)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-8027	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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